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S.510 METAL SIDING, ROOF AND INTERIOR WALL

1. SCOPE OF WORK

- 1.1 Metal siding, roof, and interior wall work, shall include the Bidder's designing, fabricating, transportation, furnishing, erecting, inspection and testing of the work together with all rain water gutters, flashing, fasteners and material necessary to ensure a watertight construction.
- 1.2 Providing penetrations through siding, walls and roofing shall be also required for the Bidder's Work.
- 1.3 This work shall conform to the requirements specified in Subsection 1,2 "Summary of Work" and S-001 "General Design Requirement of Structure, Civil and Building Works".

2. DEFINITIONS

- 2.1 Metal Siding: Exterior enclosure of one or more sides of a structure, insulated or uninsulated. If the siding is insulated, it shall consist of exterior metal sheets, insulation, subgirt and interior metal sheets (which are the siding components). If the siding is uninsulated, it shall consist of the single exterior metal sheets only. Both types of the siding shall also include all related fasteners and other accessories.
- 2.2 Metal Roof: Exterior enclosure of the roof structure, insulated or uninsulated, constructed of the same components, fasteners and other accessories as for metal siding (in contrast to metal decking covered with concrete and/or built-up roofing).

If the roofs are insulated, the components shall consist of exterior metal sheets with insulation underneath, and with interior metal sheets. If roofs are uninsulated, components shall consist of the single exterior metal sheets only.
- 2.3 Metal Interior Wall: Enclosure of one or more sides of a space within a structure, insulated or uninsulated, constructed of the same components, fasteners and other accessories for metal siding (in contrast to prefabricated metal partitions or similar constructions). If walls are insulated, components shall consist of two metal sheets enclosing and insulation core, and with or

without girts depending on the method of construction. If walls are uninsulated, components shall consist of an assembly of single metal sheets.

- 2.4 Fire Rated Insulated Fluted Metal Siding shall consist of field assembled panels utilizing a liner panel, insulation, subgirts, gypsum board and fluted extension metal sheet.

3. CODES AND STANDARDS

- 3.1 The Bidder shall comply with the requirements of Part Subsection 2.5 “Codes and Standards”.

- 3.2 The codes and standards of the following organization extra from to the Part 4, subsection 2.5 “Codes and Standards” are specifically applicable to the design, manufacture and testing of the work included in this specification:

AISI		American Iron Steel Institute
ANSI		American National Standards Institute
ASTM		American Society for Testing and Materials
NBS		National Bureau of Standards, USA
SAE		Society of Automotive Engineers,
UL		Underwriters’ Laboratories, Inc
JIS		Japanese Industrial Standards

4. DESIGN REQUIREMENTS

4.1 STRUCTURAL REQUIREMENTS

- 4.1.1 The Bidder shall be fully responsible for the design of siding, roofs and interior walls to withstand the specified loads, without at exceeding the maximum allowable unit stresses according to AISI, and without exceeding the allowable deflections specified in Clause 4.1.2. Subgirts shall be used for all insulated siding. Panels may be designed as continuous over two or more spans, according to the girt or purlin spacings covered.
- 4.1.2 The Bidder shall have qualified engineers to perform or to supervise the design work.
- 4.1.3 Fasteners for attaching siding, roofs and interior wall component to each other and to girts and other structural supports shall be designed to with-

stand the specified loads without exceeding maximum allowable unit stresses for such fasteners.

- 4.1.4 The maximum allowable deflection for insulated or uninsulated siding, roofs or interior walls subject to the design loads shall not exceed 1/180 of the clear span between girts or purlins, in the direction normal to the plane of the siding, roofs or interior wall.
- 4.1.5 Design wind loads shall conform to the applicable requirements, specified in S-002 "Loading Criteria" with wind loads based on maximum wind speed of 120km/hr.

4.2 THERMAL REQUIREMENTS

The overall coefficient of thermal transmission U for insulated siding, roofs and interior walls shall not exceed 0.75 watt/m² -°C in still air.

Test data shall be submitted to the Owner, if requested, to substantiate the Bidder's claims as to the thermal properties of his materials used for construction.

4.3 ACOUSTIC REQUIREMENTS

If acoustically treated areas are required, they shall have a Noise Reduction Coefficient (NRC), in conformance with ASTM C423 of not less than 0.85 and the Bidder shall submit a test report to the Owner for approval.

4.4 WEATHERTIGHTNESS

All completed siding and roofs shall be weather and watertight and dust tight.

Provision shall be made, in the design of insulated siding to drain to the exterior of siding of any condensation forming within the assemblies.

5. MATERIALS AND DETAILS

5.1 METAL SHEET

- 5.1.1. Material shall be hot dipped galvanized steel complying in all respects with ASTM A 653 Grade B Coating designation G 90 Zinc coated steel sheets
- 5.1.2. Metal thickness: to be fluted as required for loading and span, but not less than 0.72mm thickness
- 5.1.3. Finish shall be baked enamel on both surfaces.
- 5.1.4. Texture shall be smoothly.
- 5.1.5. Acoustic treatment: where an acoustic treatment is required for insulated siding, the interior sheets shall be perforated with holes of size and spacing under a NRC value of not less than 0.85, as specified in Clause 4.3.
- 5.1.6. Finish coating shall comprise of PVF2 Plus , double or two layer poly vinylidene fluoride applied on primer for both surfaces (exterior and interior) of insulated roof and wall panels, see table S510 -1
- 5.1.7. Color: to require the Owner's approval.

5.2 INSULATION

Where insulated siding & roof is indicated, the insulation shall conform to the following requirements:

- 5.2.1 Material: shall be fibrous glass, rock or slag with minimum density of 25.3 kg/m³.
- 5.2.2 The roof insulation shall be reinforced on down face with vinyl or aluminium foil. The thickness of vinyl shall not be less than 0.01 mm and the thickness of aluminium shall not be less than 0.025 mm
- 5.2.3 The wall insulation shall be completely enclosed with 150 micron thick polyethylene protective cover.
- 5.2.4 Thickness of insulation shall be a minimum of 3.75 cm

5.3 FIRE HAZARD CLASSIFICATION RATING

The layer of insulation, including aluminum foil facing, shall meet the following Fire Hazard Classification Rating of Underwriter's Laboratories, Inc:

- 5.3.1 Flame Spread = 10 - 15
- 5.3.2 Fuel Contributed = 0 – 5
- 5.3.3 Smoke developed = 0

5.4 FIRE RATED EXTERIOR WALLS

The materials shall be same as insulated interior metal siding except gypsum boards, as per Clause 2.4, and the thickness and number shall be determined by a two hours fire rating.

- 5.4.1 Panel thickness shall be 3.75 cm nominal overall.
- 5.4.2 Interior panel sheets shall be hot dipped galvanized interior type sheets as per Clause 5.1.

5.5 FLASHING

- 5.5.1 Exterior flashing: Material, pattern and finish shall be same as exterior sheet of siding and roofs with minimum thickness of 0.72 mm
- 5.5.2 Interior flashing: Material shall be galvanized steel with thickness and finish same as interior sheet of insulated siding and roofs.

5.5.3 Flashing shall be provided wherever required but not limited to the following:

5.5.3.1 Corners, roofs, ridges, caps, etc.

5.5.3.2 Windows, doors and louvers.

5.5.3.3 Beam penetrations, wall or roof sleeves, duct opening and miscellaneous opening, etc.

5.6 FASTENERS

5.6.1 Type of Fasteners

One of the following three types of screw fasteners shall be used as indicated for the fastening of metal siding, roofs and interior walls:

5.6.1.1 Type A: Type A sheet metal screws to have pointed ends and wide pitch threads. This type of screw is used for connecting gauge metal to gauge metal.

5.6.1.2 Type B: Type B self tapping screws to have flat ends and close pitch threads. This of screw is used for connecting gauge metal to gauge metal.

5.6.1.3 Type AB: Type AB self tapping screws to have pointed ends and close pitch threads. This type of screw is used for connecting gauge metal to a structural member.

5.6.2 Material shall be type 305 stainless steel with cadmium plated finish and the heads painted to match color of siding.

5.6.3 Size of Fasteners: No. 12 or larger

5.6.4 Length of Fasteners

5.6.4.1 The length type A screw shall be sufficient to engage gauge metal sheets together with a minimum projection of 1.0 cm beyond the sheet surface.

5.6.4.2 The length of the type B or Type AB screw shall be such as to engage a sufficient thickness of support to which they attached, to develop the full capacity of fasteners.

5.7 PARAPET CAP:

Same as exterior flashing

5.8 CAULKING & SEALING TAPE:

Manufacturer's standard

5.9 LAPS

5.9.1 Lap lines for siding shall line up all around the building.

5.9.2 End laps shall be 15 cm minimum long for wall and 25 cm minimum long for roof respectively.

5.9.3 Side laps shall not be less than ½ rib.

5.10 COATING FOR METAL SIDING AND ROOFS

Coating for metal siding and roof shall be meet the detail requirements specified in Table S-510.

5.11 RAINWATER GUTTERS

Rainwater gutters and support shall comprise of steel sheet metal and shall be protected by being hot dip galvanized with a minimum of 0.250 kg/m² zinc coating. The outer surface shall be completed with an applied protective coloured finish same as the exterior flashing to match the siding.

5.12 PENETRATIONS, WALL OR ROOF SLEEVES

- 5.12.1 Penetrations, sleeves, duct openings, etc, in siding, roofs and interior walls shall be shown on the Bidder's drawings and shall be soundly designed for weather tightness.

6. STORAGE

- 6.1 All siding materials, components, fasteners and other accessories shall be stored dry, of the ground and under protective cover, in a satisfactory manner.
- 6.2 The Bidder shall arrange with the Owner for temporary storage of materials, and shall be responsible for storing these materials in such a manner as to preclude damage, deterioration, etc. all costs for such protection shall be included in the Contract Price.

6.3 ERECTION

Erection shall be in strict accordance with manufacturer's procedure for this type of work. Care shall be taken to insure that steel and aluminium should be at all times separated such that a galvanic reaction cannot take place.

7. CLEANING OF SIDING, ROOFS AND INTERIOR WALLS

- 7.1 All exposed faces of siding, metal roofs and interior walls shall be cleaned of labels, dirt, streaks, caulking or sealing materials, undue discoloration etc. and as a minimum a dry wipe down shall be provided at the time of erection.
- 7.2 All exposed faces of siding, metal roofs and interior walls, just prior to acceptance by the Owner shall be completely satisfactory to the Owner in appearance.

8. REMEDY OF DEFECTS IN APPEARANCE

- 8.1 Defects in appearance shall include nicking, scratching, gouging, dimpling, denting, abrasions, and marring of any nature due to shipping, storing, handling, erection, weathering etc. and shall be remedied at Bidder's expense as follows:
- 8.2 Touch up the baked enamel organic coating finish with one or more coats of the same coating material used for the original finish. Thickness, appearance and color of the touch-up shall match adjacent surfaces if touch-up is not satisfactory to the Owner, damaged sheets or accessories shall be replaced.

9. PROTECTION OF INSULATION DURING ERECTION

For all insulated constructions, the Bidder shall satisfactorily protect all insulation from the weather during erection so as to eliminate soaking of insulation by water, or other moisture contaminants.

10. INSPECTION AND TESTING

The Bidder shall be responsible for all inspection and testing required by this Specifications unless otherwise specified. The Owner may during the course of the work, observe various phases of the work performed by the Bidder, at the Bidder's plant and at the site.

Table S-510 Organic Coating for Metal Siding, Roofs and Accessories

Item	
1.Base Metal	Material : Galvanized steel, ASTM A653, G90 Coating
2.Coating	2.1 Barrier Coat: Manufacturers standards 2.2 Prime Coat : Epoxy coating 2.3 Finish coat: liquid formulation based on PVF2 Plus Kynar 500 fluorocarbon resin or products meeting the same requirements; consisting of the least 80 % 2.4 Dry Film Thickness: 2.4.1 Barrier Coat : As per manufacturer's standard. 2.4.2 Prime coat: (front side) 6 microns ~ 8micronsa 2.4.3 Finish Coat : (front side) 40 ±5 microns (reverse side) 10± 15 microns 2.4.4 Total Coating: Not less than 62 microns
3.Curing	Oven baked in accordance with manufacturer's standard.
4. Tests	Tests shall conform to, or exceed the following tests requirements: 4.1 Pencil Hardness (ASTM D3363) : F minimum 4.2 Salt Spray (ASTM B117)after testing 600 hours in 5 % salt fog at 35⁰C, samples area shall have no blisters. 4.3 Humidity (ASTM D2247): After 1000 hours testing at 100% relative humidity and 38°C, samples shall have no blisters. 4.4 Weatherometer (ASTM G23): after 5000 hours exposure in Sunshine Arc Weatherometer or Atlas Twin Weatherometer, no chalk in excess of No.8 per ASTM D659 and no color fade more than 5 NBS units per ASTM D2244, as operated under Dew Cycle conditions using 60/60 cycle. 4.5 Flexibility (ASTM D1737) : Withstand 1T 1.5 T bends

Item	
	4.6 Gloss (ASTM D523) :25 -35 % reflectivity measured on 60⁰ glossmeter. 4.7 Impact : No removal of finish when tested as follows: 4.7.1 Using a sharp knife, make 6 or, more parallel cuts at 1.5 mm intervals through finish to bare metal. Cross hatch similarly. 4.7.2 Reserve impact to at least 92 cm-kg or to point of metal rupture. 4.7.3 Apply Scotch cellophane tape firmly to scribed impacted area. Remove tape with sharp pull. There shall be no loss of adhesion. 4.8 Abrasion (ASTM D968): the average coefficient of abrasion shall be not less than 282 litres lowest with the value not less than 200 litres.
5. Warranty	The Bidder shall furnish the Owner ,upon acceptance installation, a written warranty that fluorocarbon organic coating has been applied in accordance with licensee manufacturer's process instructions and under normal conditions; and that it shall meet the following standards for a minimum period of 10 years (the Bidder shall state the number of years warranty, and also the number of times he will replace any defective panels without charges of any kind): 5.1 Will not chalk in excess of ASTM D659 No.8 rating. Chalk rating shall be determined by the procedure outlined in ASTM D659. 5.2 Will not change color more than 5 NBS units as determined in accordance with procedures set forth in ASTM D2244, after removal of external deposits and chalk. 5.3 Will not crack, check or peel (lose adhesion).

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